Saliency-Guided Transformer Attention With Pixel-Level Contrastive Learning for Weakly Supervised Defect Localization

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***Abstract*—Weakly supervised learning has emerged as a powerful paradigm for image segmentation, providing a practical solution to reduce the dependence on costly, pixel-level annotated datasets. By leveraging partial annotations, such as image-level tags, this approach significantly reduces the labeling burden while maintaining competitive performance. However, weakly supervised methods inherently face challenges, including overlapping activation regions and insufficient localization due to sparse and noisy signals from image-level tags, often leading to suboptimal segmentation performance. These issues are further exacerbated in industrial defect localization, where datasets present unique complexities, including low-contrast object boundaries, inconsistent shapes, interclass similarities, and substantial intraclass variations. To address these limitations, we introduce a novel saliency-guided transformer attention with contrastive learning framework. The proposed framework leverages Transformer attention to generate localization maps and enhances the learning of challenging foreground and background information through saliency-guided cues. In addition, a pixel-level contrastive learning module is employed to refine feature map representations by bringing positive pairs closer together and pushing negative pairs apart, effectively addressing challenges, such as overlapping activations and ambiguous boundaries. Through extensive experiments and ablation studies, the proposed method demonstrates superior performance compared to state-of-the-art approaches across three defect segmentation datasets. We also evaluated the generalization of our approach on the PASCAL VOC segmentation and MVTec anomaly detection datasets.**

***Index Terms*—Class-specific class activation map (CAM), contrastive learning, defect localization and segmentation, transformer attention, weakly supervised learning.**

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The implementation code for this study is available online. Code is available online at https://github.com/djene-mengistu/STAC.

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I. Introduction

In recent years, deep learning-based solutions for visual tasks have emerged as one of the most researched and applied advancements in the field of artificial intelligence (AI). Within intelligent manufacturing, the rapid progress in AI algorithms, Big Data, and advanced computing technologies has positioned deep learning as a leading research area. These advancements serve as key enablers driving the future of manufacturing automation, including improved productivity, enhanced product quality, greater flexibility, and reduced operating costs [1]. One prominent application of deep learning in manufacturing is the automation of machine vision processes, enabling industrial machines and equipment to interpret visual information for tasks, such as inspection, guidance, process control, and more. Automated visual inspection, in particular, plays a vital role in modern manufacturing. While traditional methods for industrial defect inspection have been used for decades, they are often error-prone, inflexible, and expensive. In contrast, deep learning-based systems provide an efficient, accurate, and adaptable alternative that easily accommodates the dynamic changes in the manufacturing settings [1], [2]. Despite these advantages, the application of deep learning-based methods to industrial defect inspection is still a relatively recent development and remains a highly active area of research. One of the primary challenges lies in the reliance of these solutions on annotated datasets, making it difficult to develop generalized and scalable approaches, especially for tasks requiring precise defect localization and analysis, such as defect segmentation. Consequently, recent research works have shifted toward label-efficient methods, including semisupervised [2], [3], [4], [5] and

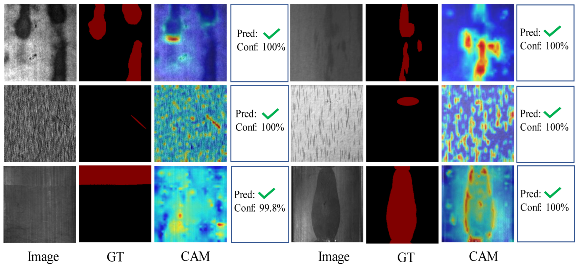

Fig. 1. Challenges in weakly supervised segmentation for industrial images. Despite high-confidence classification, the localization heatmap fails to accurately highlight defect regions, exposing a misalignment between classification decisions and spatial attention. This underscores the limited explainability of such models, where accurate classification does not guarantee true learning of relevant features.

weakly supervised learning approaches [6], [7], [8], [9], [10], [11], [12], [13].

Weakly supervised segmentation relies on partial annotations, such as image-level tags, bounding boxes, and scribble annotations, to achieve precise object localization, primarily using class activation maps (CAM) to highlight highly activated regions in an image. Although there is a common challenge of a supervision gap between classification and segmentation models, this approach has demonstrated remarkable performance in common object segmentation, achieving precise object boundaries and localization map. However, unlike common object segmentation datasets, industrial defect data exhibit high variability in shape, object boundaries, appearance, and contrast [2], [6], [7]. This variability makes it difficult for CAM-based methods to accurately localize and delineate object boundaries. In addition, inconsistencies in background–foreground contrast, shape, and appearance lead to substantial imbalances, such as interclass differences and intraclass similarities. These challenges exacerbate the co-occurrence problem in weakly supervised segmentation, where the same regions may be activated for different classes due to overlapping features, further complicating precise localization [6], [7], [14]. For instance, as shown in Fig. 1, the classification network activates irrelevant regions despite achieving near-perfect classification confidence. Even when the classification result is correct, this does not guarantee that the model accurately identifies the underlying defect regions, further highlighting the challenges in weakly supervised segmentation of industrial images.

While several methods have been proposed to improve CAM, including multitask learning [15], [16], self-supervision [17], [18], and Transformer attention mechanisms [19], [20], [21], their effectiveness has been mainly demonstrated on natural image datasets, such as COCO and Pascal-VOC [14]. Industrial images, however, introduce additional challenges that make it difficult for these methods to achieve precise defect localization [22].

Addressing these challenges, this article proposes a novel saliency-guided Transformer attention-based CAM framework with contrastive learning for weakly supervised defect localization and segmentation. The proposed method leverages Transformer attention guided by class-agnostic saliency cues

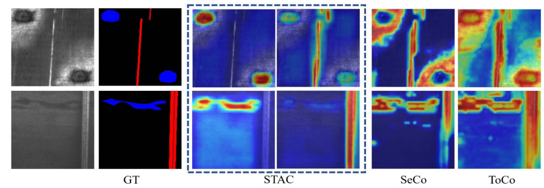

Fig. 2. Illustration of the class-specific localization capability of the proposed method (STAC) versus selected methods.

to enhance foreground–background separation in challenging industrial images with low contrast and complex backgrounds. In addition, a pixel-level contrastive learning module (PCM) is introduced to improve feature discriminability by reducing interclass confusion and handling intraclass variability in defect regions (see Fig. 2). Transformers are well suited due to their ability to capture global context through self-attention, enabling better localization of fine-grained and ambiguous defects. Unlike CNN-based CAMs that often focus on limited discriminative regions, Transformer attention maps provide richer spatial cues without requiring architectural modifications or additional passes. The contributions of the study are summarized as follows.

1) We propose a novel weakly supervised learning framework that leverages Transformer attention to generate a localization map, which is then enhanced by saliency cues and pixel-level contrastive learning for improved defect localization and segmentation.
2) We present an effective framework for generating class-to-patch attention, which produces class-specific localization maps, by leveraging the pairwise attention maps of a Transformer. This attention is further enhanced by a CNN-based auxiliary CAM refinement module, which subsequently enhances foreground and background learning via saliency cues.
3) We propose a PCM that enhances feature representation by minimizing intraclass variance and maximizing interclass separation, using positive and negative pairs derived from progressively refined CAMs.
4) We validated our approach on three different defect segmentation datasets: NEU-Seg [23], DAGM [24], and MTD [25], and demonstrated that the proposed method resulted in state-of-the-art performance compared to several methods. Furthermore, we evaluated our method's generalization across domains on the common object segmentation with PASCAL VOC2012 [26] dataset and on anomaly detection with the MVTec [27] dataset.

II. Related Works

A. Weakly Supervised Semantic Segmentation (WSSS)

WSSS aims to achieve accurate pixel-level classification in semantic segmentation tasks using partial annotations, such as image-level labels, bounding boxes, saliency cues, or scribbles. Unlike fully supervised methods that require precise pixel-level annotations, WSSS relies on cost-effective and accessible annotations while maintaining competitive accuracy. This makes it a practical solution for large-scale image segmentation tasks, particularly in areas, such as medical image analysis and defect segmentation, where pixel-level training samples are scarce or expensive to obtain [14], [28].

1) CAM-Based Methods: The most commonly used approach in WSSS involves utilizing image-level tags to generate CAMs, which provide coarse localization based on the most activated regions in an image classification network. These generated CAMs are then employed as pseudolabels to train a segmentation network. However, since the generated CAMs often fail to capture precise object boundaries, their performance is typically suboptimal in providing effective supervision signals. As a result, various methods have been proposed to enhance the quality of CAMs, including multitask learning [15], [16], contrastive representation learning [18], [29], [30], [31], consistency under image transformations [17], [31], self-supervised learning [17], [18], [32], [33], and cross-image semantic information learning [30], [34].

AuxSNet [16] employs an auxiliary multitask learning approach that integrates multilabel classification and saliency detection with the primary segmentation task, while EPS [15] proposed explicit pseudosupervision using image-level labels combined with saliency cues to optimize the object boundaries and localization maps of CAMs. ACR [35] employs an adversarial learning framework, pitting an image classifier against an image reconstructor to promote multitask learning and develop richer representations, which enhance localization performance. For the defect inspection task, FB-SAM [11] extended the CAM-based approach by introducing a localization method enhanced with spatial attention association to better focus on small defects. Similarly, CSS [13] proposed a category-agnostic strategy for surface defect localization by clustering features and selectively suppressing nondefect activations, while CAAWSL-Net [12] further refined the CAM-based framework through background-constrained training in a multistage setting. In addition, building upon the initial CAM seeds, several approaches have employed deep level set learning and shape priors to enhance segmentation accuracy [8], [10], [36]. However, these methods still rely on CAM, which struggles to accurately localize challenging and complex defects characterized by low contrast and ambiguous boundaries.

Self-supervised and contrastive learning methods have also been proposed to enforce consistency across different views and scales, ensuring that CAMs generated from various transformed versions of the image remain aligned [17], [18], [31], [32]. Through spatial transformation consistency, this approach encourages the model to learn enhanced representations by capturing spatial, semantic, and contextual relationships. Self-supervised equivariant attention mechanism (SEAM) [17] applied consistency on the CAMs generated from various transformed images for supervision signal in a self-supervised setting, while sub-category exploration (SCE) [32] introduces self-supervised dataset-level image feature clustering to generate subcategories for each class and encouraging the network to perform more challenging tasks beyond focusing on the most discriminative object part. P2P [31] proposed cross-view feature semantic consistency across different views of an image,

TABLE I
COMPARISON OF THE MIOU METRICS OF THE GENERATED CAM WITH THE PROPOSED METHOD AND OTHER SOTA METHODS

Methods	Backbone	NEU-Seg		DAGM		MTD	
		Train	Test	Train	Test	Train	Test
EPS [15]	Res38	69.62	68.25	56.67	54.32	45.34	46.59
SEAM [17]	Res38	52.68	52.12	45.19	44.50	38.30	38.86
P2P [31]	Res38	55.86	54.42	35.15	34.84	35.71	37.05
L2G [18]	Res38	65.78	64.32	55.21	54.19	42.75	43.02
T-CAM [37]	Conf-S	53.97	53.42	48.74	50.06	35.68	34.94
AuxSNet [16]	Res38	65.02	63.64	46.22	47.06	40.86	41.84
MCT+ [19]	DeiT-S	43.57	42.67	46.63	46.35	28.12	28.79
SeCo [21]	ViT-B	46.34	49.14	56.20	57.71	33.62	33.69
ToCo [20]	DeiT-B	47.13	45.94	53.30	54.80	28.23	27.56
CTFA [38]	DeiT-B	53.00	52.88	54.87	56.07	28.57	29.59
CTI [34]	DeiT-S	52.07	51.01	46.82	46.67	41.22	39.65
STAC	DeiT-S	**75.70**	**74.71**	**65.47**	**66.53**	**67.60**	**67.17**

whereas L2G [18] proposed consistency of attention between multiple randomly cropped local patches of an image with the global image attention, enhancing local-to-global knowledge transfer for better localization of CAMs.

2) Transformer Attention-Based Methods: Transformer networks have become a popular approach for image recognition tasks in recent years. By employing self-attention mechanisms, Transformers model long-range dependencies by dynamically assigning weights to different parts of the input sequence, enabling a comprehensive understanding of the global context. This attention mechanism can be adapted to generate CAMs by associating attention scores with spatial regions, effectively highlighting areas relevant to specific classes. These CAMs can then be refined and utilized as pseudolabels for semantic segmentation, facilitating precise region localization without requiring pixel-level annotations. Consequently, numerous methods have been proposed to enhance object region localization in WSSS settings by leveraging the powerful attention mechanisms of Transformer networks [19], [20], [21], [34], [37], [38].

T-CAM [37] applied attention weights from the Transformer network to explicitly refine the CAMs generated by a CNN branch. MCT+ [19] applies multiclass tokens with Transformer attention to achieve class-specific localization of attention maps, encouraging learning interactions between the class tokens and patch tokens. ToCo [20] introduces class token contrastive learning to enforce consistency between class tokens of local image patches and the global image, where SeCo [21] introduces a method to decouple co-context in the image space to enhance class-specific representations and address co-occurrence or overlapping activation issues. Extending on the concept of ToCo [20], CTFA [38] applied auxiliary decoders to improve activation of the foreground areas in remote sensing scenes.

Although existing methods have shown strong performance in weakly supervised segmentation of common objects with consistent boundaries and regions, as reported on datasets, such as COCO and PASCAL VOC, challenges, such as overlapping activations and imprecise localization stemming from the sparsity of image-level labels, remain significant and are still an active area of improvement in recent research works [14], [21], [34]. These limitations make the application of such methods particularly challenging in industrial defect localization and segmentation. Addressing these issues requires the development

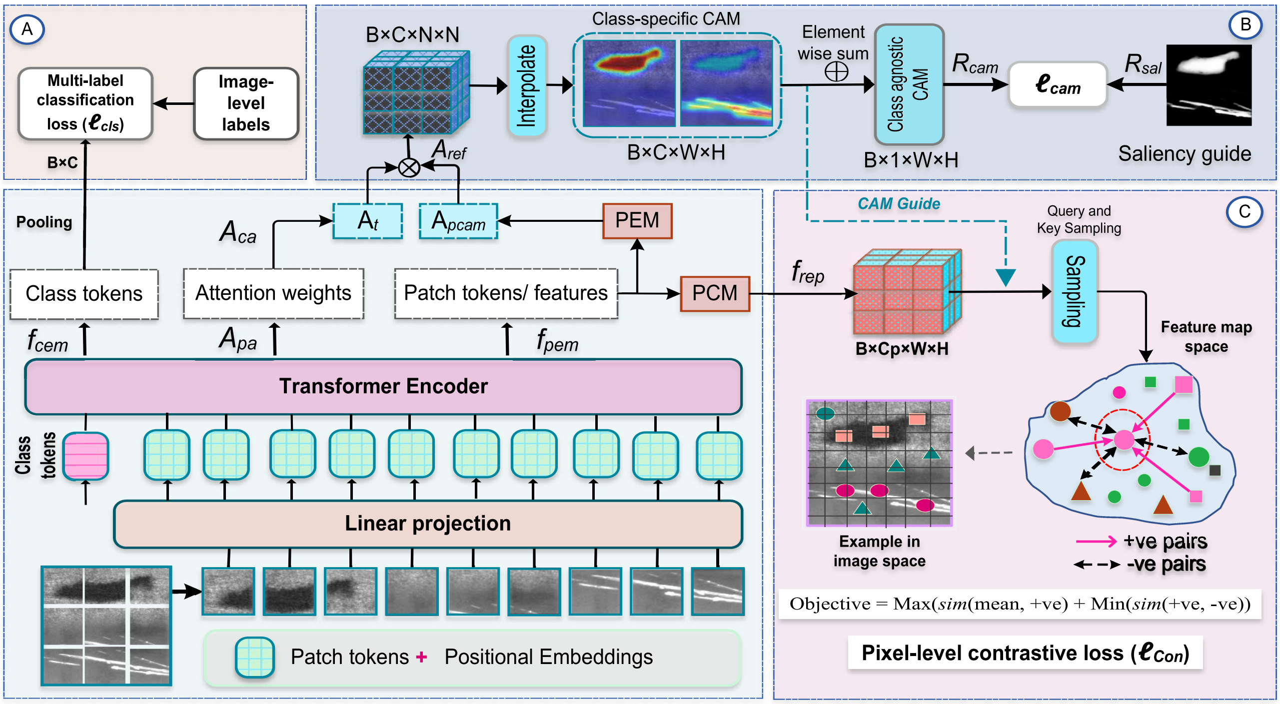

Fig. 3. Overall framework of the proposed method. The Transformer network is trained using image-level labels to generate attention maps (A_{pa}) and Token features (f_{cem} and f_{pem}). The attention map A_{ca} is extracted and refined with auxiliary CAM (A_{pcam}) from PEM module to generate A_{ref}. The saliency map then guides the localization of the refined attention map. The refined CAMs then guide the selection of positive and negative pairs for pixel-level contrastive learning using the representations computed from the PCM module. The framework consists of: (A) multilabel classification loss, (B) saliency-guided CAM refinement module, and (C) CAM-guided PCM.

of tailored approaches that account for these complexities to achieve accurate defect localization and segmentation.

III. Methods

A. Overview of the Proposed Method

In this study, we propose a novel saliency-guided transformer attention framework with pixel-level contrastive learning for defect localization and segmentation. Our approach leverages a Transformer network to generate class-specific CAMs from attention maps. These CAMs are further refined through a series of enhancement techniques to produce high-quality localization maps and segmentation pseudolabels, which are subsequently used to train segmentation networks. The proposed framework consists of three key components: a Transformer encoder block, a CNN-based CAM generation module, and a contrastive learning module, as depicted in Fig. 3. An image is first divided into $N \times N$ patches, which are then vectorized and projected into patch tokens ($f_p \in \mathbb{R}^{M \times D_e}$), where D_e is the embedding dimension and $M = N^2$. In addition, C class tokens ($f_c \in \mathbb{R}^{C \times D_e}$) are appended for each label in the multiclass task. These inputs, in addition to the positional embeddings, are fed to the Transformer encoder network. The overall shape of the input is $F \in \mathbb{R}^{(C+M)\times D_e}$. The transformer encoder consists of multihead attention (MHA), batch-norm, and multilayer perceptron (MLP). In this article, we adapt the standard configuration of the DeiT [39] Transformer. In Transformer encoder, the input is projected into three sequences of vectors representing query, $Q \in \mathbb{R}^{(C+M)\times D_e}$, key, $K \in \mathbb{R}^{(C+M)\times D_e}$, and value, $V \in \mathbb{R}^{(C+M)\times D_e}$. Using the MHA, standard scaled dots product attention is applied to calculate the attention score between all elements of the query and key sequences (1). By aggregating the attention weights from all the token sequence, each token will be updated dynamically

$$A(Q, K, V) = \text{Softmax}\left(Q.K^T/\sqrt{D}\right).V. \tag{1}$$

The computation yields a pairwise attention map, A_{pa}, with shape $\mathbb{R}^{(C+M)\times(C+M)}$, which represents the interactions between each token. In addition, the Transformer encoder produces the feature map embeddings, $\mathbb{R}^{(C+M)\times D_e}$, which are subsequently processed by the auxiliary CNN-based CAM module and the contrastive learning module.

B. Class-Specific CAM Generation and Refinement

After computing the pairwise attention map in (1), A_{pa}, the class attentions to patches, A_{ca}, is extracted as $A_{\text{ca}} = A_{\text{pa}}[1 : C, C+1 : C+M]$, resulting in $A_{\text{ca}} \in \mathbb{R}^{C\times M}$, where values represent the attention score of each class to patches. This attention map is computed across all Transformer layers (L). The final class-specific attention map is computed by aggregating the attention maps from different layers, with the impact of aggregation depth is evaluated in the ablation study (see Section IV-E)

$$A_t = 1/L \sum_{\ell}^{L} A_{\text{ca}}^{\ell} \tag{2}$$

where the resulting attention is $A_t \in \mathbb{R}^{C\times N\times N}$.

C. Patch Embedding-Based Refinement Module (PEM)

The Transformer encoder outputs patch token feature map embeddings, $F_{\text{em}} \in \mathbb{R}^{(C+M)\times D_e}$, from which class embeddings, $f_{\text{cem}} \in \mathbb{R}^{C\times D_e}$, and patch embeddings, $f_{\text{pem}} \in \mathbb{R}^{M\times D_e}$, are extracted. The class token embeddings are passed to global average pooling (GAP) for multilabel classification, while the patch token feature maps are processed by an auxiliary CNN-based CAM module (PEM) and a projection head for contrastive learning. The PEM module processes input of shape $\mathbb{R}^{M\times D_e}$ to produce class-specific CAM outputs, $A_{\text{pcam}} \in \mathbb{R}^{C\times N\times N}$, where C is the number of classes. This CAM refines the Transformer attention map from (2) through elementwise multiplication, as shown in (3). The refined attention map (A_{ref}) is then evaluated against precomputed saliency cues to sharpen its focus on relevant foreground objects (defects), while the multilabel classification guides the identification of specific classes within the foreground (see Section III-E-2)

$$A_{\text{ref}} = A_t . A_{\text{pcam}}. \tag{3}$$

D. Pixel-Level Contrastive Learning Module

The PCM takes the Transformer patch embeddings, $f_{\text{pem}} \in \mathbb{R}^{M\times D_e}$, as input. The PCM module processes these patch token feature embeddings through a series of operations: a 3×3 convolution, batch normalization, rectified linear unit (ReLU) activation, a 1×1 convolution, and upsampling. This transformation produces feature representations, $f_{\text{rep}} \in \mathbb{R}^{C_p\times W\times H}$, where C_p denotes the projection channel dimension, which is set to 256, and W and H correspond to the input image dimensions. The resulting f_{rep} is then used to sample positive and negative pairs for computing contrastive loss (see Section III-E-3).

E. Training Process

1) Multilabel Classification Loss: The network is trained using the standard multiclass classification loss with image-level labels, enabling it to detect the presence of specific defect classes in an image, while the attention map highlights the regions that are likely to contain the described class of defects. We used multilabel soft margin loss (4) for classification. GAP is applied to the class token embeddings, transforming $\mathbb{R}^{C\times D_e} \xleftarrow{\text{GAP}}$ to $y_{\text{cls}} = \mathbb{R}^{C}$

$$\ell_{\text{cls}} = -\frac{1}{C}\sum_{i=1}^{C} y^i \log \alpha\left(y_{\text{cls}}^i\right) + \left(1-y^i\right)\log\left(1-\alpha\left(y_{\text{cls}}^i\right)\right). \tag{4}$$

2) Saliency-Guided CAM Refinement Loss: For the saliency guidance, we first compute saliency cues from pretrained and fine-tuned PFAN [40] for each dataset. To align with the original input image dimensions, the refined CAM, A_{ref}, is upsampled using linear interpolation, resulting in a class-specific map of shape $\mathbb{R}^{C\times W\times H}$. We then applied normalization and elementwise sum across the class dimension to produce a class-agnostic attention map, R_{cam}, with shape $\mathbb{R}^{1\times W\times H}$, as shown in Fig. 3, part B, for comparison with the precomputed saliency map, R_{sal}. This process yields total activation strength at each pixel location, with values approaching 1 for foreground regions and 0 for background regions. To evaluate the alignment between R_{cam} and R_{sal}, we compute MSE loss, as in (5). When the saliency map highlights a foreground region, the summation prioritizes contributions from the foreground class associated with the defect in the image, while suppressing activations from other foreground classes. This behavior is enforced through gradient updates guided by the image-level label, which indicates the presence of specific defect classes. Consequently, the saliency map facilitates foreground–background separation, while the classification network ensures that attention is focused on the correct foreground class over the training iterations

$$\begin{aligned} A_{\text{fg}} &= \frac{A_{\text{ref}} - \min_{h,w} A_{\text{ref}}}{\max_{h,w} A_{\text{ref}} - \min_{h,w} A_{\text{ref}} + \epsilon} \\ R_{\text{cam}} &= \sum_c A_{\text{fg}}^{(b,c,h,w)} \in \mathbb{R}^{B\times 1\times H\times W} \\ \ell_{\text{cam}} &= \text{F.mse}(R_{\text{cam}}, R_{\text{sal}}). \end{aligned} \tag{5}$$

3) CAM-Guided Pixel-Level Contrastive Loss: In pixel-level contrastive learning, the selection of positive and negative pairs is guided by the refined CAM (see Fig. 3). To minimize the impact of noisy or uncertain predictions, we apply thresholding on these CAMs to derive pseudopixel-level labels, selecting high-confidence pixels for contrastive learning. This ensures that the most reliable regions are used for training and dynamically adapt to CAM quality as the training progress

$$\begin{aligned} A_{\text{fg}} &= \frac{A_{\text{ref}} - \min_{h,w} A_{\text{ref}}}{\max_{h,w} A_{\text{ref}} - \min_{h,w} A_{\text{ref}} + \epsilon} \odot \mathbf{T} \\ P_{\text{bcg}} &= \mathbf{1} - \max_c A_{\text{fg}} \\ P &= P_{\text{bcg}} \oplus A_{\text{fg}} \in \mathbb{R}^{B\times(C+1)\times H\times W} \\ P_{\text{pseudo}} &= \arg\max_c P \in \{0,1\}^{B\times(C+1)\times H\times W} \\ M &= \mathbf{1}\left[\max_c P^{(b,c,h,w)} \geq \eta\right] \end{aligned} \tag{6}$$

where $\mathbf{T} \in 0, 1^{B\times C}$: target labels for present classes, $\odot$: elementwise multiplication with broadcasting, P_{bcg}: background probability, ϵ: small constant for numerical stability, $\oplus$: concatenation of channels, and η: confidence threshold for reliable pixels (set to 0.6). The computed pseudolabels (P_{pseudo}) and mask (M) will then guide the selection of positive and negative keys from the representations (f_{rep}) for contrastive learning.

The contrastive loss encourages queries p_q to be similar to the positive keys p_k^+, and dissimilar to the negative keys p_k^-, bringing the related feature maps together and pushing the unrelated (different class) features away from each other. The queries and keys are sampled from the projection module output, which is explained in Section III-D: $p_q, p_k^{+,-} \in \mathcal{P}$, where details and impacts of different sampling strategy are analyzed in ablation study (see Section IV-E-4). Our implementation of the contrastive loss is inspired by the implementation of Liu et al. [41], and the overall formulation of the contrastive loss, ℓ_{Con}, is presented as follows:

$$\begin{aligned} \ell_{\text{Con}} = \sum_{c\in C}\sum_{p_q\sim\mathcal{P}_q^c} \\ -\log\frac{\exp(p_q\cdot p_k^{c,+}/\tau)}{\exp(p_q\cdot p_k^{c,+}/\tau) + \sum_{p_k^-\sim\mathcal{P}_k^c}\exp(p_q\cdot p_k^-/\tau)} \end{aligned} \tag{7}$$

Algorithm 1: Saliency-guided Transformer attention with pixel-level contrastive learning (STAC).

Input: $(x_i, y_i) \in X$, where y_i is image-level labels
Output: Network parameters θ
Initialize the network parameters θ and hyperparameters λ, γ
while *stopping criterion not met* **do**
  for *epoch = 1, 2, ..., max_epoch* **do**
    1) Sample mini-batch b of (x_i, y_i);
    2) Train the classification network using (4) (ℓ_{cls}) and extract the attention weights, A_{pa} and patch token features, f_{pem};
    3) Extract the class-to-patch attention, A_{ca}, from A_{pa} and aggregate as in (2);
    4) Using PEM module, compute A_{pcam} from f_{pem} and refine the attention as in (3);
    5) Convert the A_{ref} into class-agnostic map (5);
    6) Using the class-agnostic A_{ref} and saliency, compute the loss (ℓ_{cam}) as in (5).
    7) Using PCM module, compute the representation map f_{rep} from the f_{pem};
    8) Compute pseudo labels and mask to guide Query and Key selection (6);
    9) Sample Query and Key using (8) and (9);
    10) Compute pixel-level contrastive loss (ℓ_{Con}) using (7);
    11) Compute the total loss (10);
    12) $\mathcal{L} = \ell_{cls} + \lambda\ell_{cam} + \gamma\ell_{Con}$;
    13) Compute the gradients and update the weights: $\theta \leftarrow \theta - \eta\nabla_\theta\mathcal{L}$;
  end
  return θ
end

where C is a set with classes present in the mini-batch, τ is the softness control temperature value, $\mathcal{P}_q^c$ denotes a query set comprising all representations related with class c, $\mathcal{P}_k^c$ denotes a negative key set consisting of all representations whose labels do not correspond to class c, and $p_k^{c,+}$ denotes the positive key with mean representation of class c. Assuming that δ is a set of all pixel representations with the same resolution as $\mathcal{P}$, the queries and keys are computed as follows:

$$\mathcal{P}_q^c = \bigcup_{[i,j]\in\delta} \mathbf{1}\left(y_{[i,j]} = c\right)\, p_{[i,j]} \tag{8}$$

$$\mathcal{P}_k^c = \bigcup_{[i,j]\in\delta} \mathbf{1}\left(y_{[i,j]} \neq c\right)\, p_{[i,j]}. \tag{9}$$

The mean representation of the positive keys are represented as $p_k^{c,+} = 1/|\mathcal{P}_q^c| \sum p_q$, where $\forall p_q \in \mathcal{P}_q^c$.

The total loss for training the proposed method is given as follows:

$$\mathcal{L} = \ell_{\text{cls}} \;+\; \lambda\ell_{\text{cam}} \;+\; \gamma\ell_{\text{Con}} \tag{10}$$

where λ and γ are the coefficients to balance the loss contributions to the overall objective function. The overall training process of the proposed method is given in the pseudoalgorithm 1.

IV. Experimental Details and Results

A. Dataset Details

We conducted experiments on three defect segmentation datasets to validate our proposed method. The NEU-Seg [23] dataset consists of 3630 images of steel strip surface defects for training and 840 test samples. The second dataset, DAGM [24], is a commonly used benchmark for industrial defect detection. We selected six defect classes from this dataset, comprising 1080 training samples and 270 test samples. The third dataset, MTD [25], focuses on magnetic tile defects with five defect types. To increase the number of samples in MTD, we applied various affine transformations, expanding it to 980 training samples and 190 test samples. We employed standard preprocessing and data augmentation techniques in our experiments.

For the PASCAL VOC2012 dataset [26], we used 10 582 training samples and reported results on the 1449 validation images. Similarly, for the MVTec anomaly detection dataset, we selected five representative categories (Bottle, Carpet, Grid, Hazelnut, and Leather) for experimentation. Since the training set contains only normal images, we added augmented defective samples generated based on the test set through different spatial transformations and added them to the training set for each category, creating a training set of 600 images for each category. The results were then reported on the original defective samples. Further details on dataset preprocessing and implementation are available in the accompanying code.

B. Implementation Details

1) Backbone Networks: We used the DeiT [39] Transformer as a backbone for our framework. For the segmentation tasks, we implemented DeepLabV3+ [42], a CNN architecture, and SegFormer [43], a Transformer-based architecture. We evaluated the performance of the segmentation models using various backbone networks, including Wide-ResNet [44], ResNet [45], and EfficientNet [46]. For computing the saliency map for the training samples, we fine-tuned the pretrained PFAN [40] model and iteratively improved the consistency of the generated saliency maps using conditional random fields and postprocessing. Other saliency-detection methods [47] can also be used to improve the quality of the saliency cues.

2) SOTA Methods: We compare our method with several recent state-of-the-art WSSS methods. Among the methods are EPS [15], SEAM [17], AuxSNet [16], P2P [31], L2G [18], T-CAM [37], MCT+[19], CTI [34], ToCo [20], CTFA [38], and SeCo [21]. The details of these methods are summarized in Section II.

3) Training Details: All the experiments are implemented on NVIDIA GeForce RTX 4090 using the PyTorch deep learning framework. We trained the networks for 100 epochs using an Adam optimizer with a base learning rate of 1×10^{-3} for defect localization and 1×10^{-4} for PASCAL VOC with the cosine annealing method for learning rate scheduling. We used a batch size of 16 with standard random data augmentations and resized all samples to 224×224 for defect localization and a batch size of 4 with all samples randomly resized and cropped to

448×448 for the PASCAL VOC dataset. The loss balancing hyperparameter, λ and γ, are set to 1.5 and 0.2, respectively (see Section IV-E-5). For the SOTA methods, we followed the original implementation settings, except for the data loader pipeline, which was improved to better align with defect segmentation datasets. Further implementation details are provided in the accompanying code.

C. Evaluation Metrics

Following the commonly used metrics, we report the results on CAM and segmentation maps using mean intersection over union (mIoU). In addition, we report the false positive (FP) and false negative (FN) rates for detailed analysis of the CAM results. The mIoU evaluation is computed at different scales for better results. Following previous implementation in [19], [21], and [34], we set the scales to [1.0, 0.5, 1.5, 2.0] for testing the CAM results.

D. Results and Discussion

1) CAM Seed Result Analysis and Discussion: The WSSS method primarily aims to generate high-quality pseudolabels from image-level tags. To evaluate the quality of the generated CAM, we evaluated the proposed method and other SOTA methods using the mIoU metric. As given in Table I, the proposed method consistently outperformed other approaches across the three datasets. Despite most models achieving a classification score of over 99%, their localization accuracy (mIoU) is significantly lower. This highlights the challenges of the industrial image dataset, where models tend to focus on nonrelevant areas for classification due to substantial variations in the images. In contrast, our proposed method demonstrates superior localization of specific defect regions, attributed to the integration of saliency cues and contrastive learning that help the model to learn discriminative features for better localization. Specifically, on the MTD and DAGM datasets, most compared methods performed poorly because the backgrounds and textures in these datasets are highly variable and inconsistent, making class-specific localization more challenging only using the image level tags. The CAM seed generated by the proposed approach achieved an mIoU score of 74.71% on the test set of the NEU-Seg dataset, while the next best methods, EPS [15] and AuxSNet [16], achieved mIoU scores of 68.25% and 63.64%, respectively. Similarly, on the MTD and DAGM datasets, our proposed method significantly outperformed other approaches, achieving mIoU scores of 67.17% and 66.53%, respectively.

Despite the strong performance of Transformer-based models on large image datasets, such as PASCAL VOC and COCO, their performance in defect localization, as given in Table I, is not significantly better than other methods. This suggests that the pretrained weights provide limited guidance for object localization, unlike in common object images, which have consistent shapes, boundaries, and appearances. The multitask-based CAM refinement approaches, such as EPS [15] and AuxSNet [16], have shown better results, primarily due to the incorporation of saliency cues. In addition, self-supervised learning methods, such as SEAM [17] and P2P [31], have also demonstrated strong performance on the NEU-Seg dataset. However, as dataset variability increases, as seen in the DAGM and MTD datasets, the performance of most methods declines significantly. In contrast, our proposed method consistently outperformed others, demonstrating the effectiveness of incorporating Transformer attention, saliency cues, and contrastive learning in guiding challenging localization tasks.

To further assess the generalization capability of the proposed model across different domains, we conducted experiments on the PASCAL VOC2012 common object segmentation dataset. As presented in Table II, our method achieves a notable improvement over the baseline. As originally motivated in this study, although integrating saliency maps and contrastive learning brings some performance gains, the improvement is relatively limited compared to MCT+[19]. This is because the base model, pretrained on common objects, already exhibits strong localization ability. Consequently, the additional benefits are less pronounced on this dataset than in defect segmentation tasks, highlighting the necessity of a tailored approach for industrial defect localization.

Furthermore, the proposed method was evaluated on the MVTec anomaly detection task, as given in Table III, including implementations using CNN-based CAM generation. The results indicate that while both the CNN-based and Transformer-based baseline methods perform well, the proposed approach consistently achieves substantially better performance across all selected defect categories in the MVTec dataset. It is also worth noting that since the training involves only binary classification (defective versus normal), the baseline already performs relatively well compared to other defect localization datasets.

2) Segmentation Results: To further validate the proposed method, we trained segmentation networks using pseudolabels derived from the refined CAM. The results, given in Tables IV and V, demonstrate that these pseudolabels effectively supervised the segmentation models. Notably, the pseudolabel supervision achieved performance comparable to the fully supervised DeepLabV3+[42] with ResNet [45], as indicated in the top row of Table IV. This demonstrates the effectiveness of the proposed method in enabling annotation-efficient training and addressing the challenges of pixel-level annotation in industrial defect segmentation. Furthermore, we evaluated the segmentation performance against other weakly supervised learning methods, specifically those using level set learning for refinement, WDLS [8] and PLDL [10]. Although the results are reported from the original papers, the segmentation methods trained with the refined CAM significantly outperformed these methods.

3) Visualization of Localization Results: To further illustrate the effectiveness of the approach, we present visualizations of the generated localization map. As shown in Figs. 4 and 5, the proposed method generated consistent localization maps, even in images with challenging boundaries and contrast. Furthermore, the visualizations of the results on the PASCAL VOC and MVTec datasets are presented in Figs. 6 and 7, respectively. As shown in Fig. 7, the localization produced by the Transformer-based model is more consistent with the background, whereas

TABLE II
PERFORMANCE COMPARISON OF THE GENERATED CAM SEED ON PASCAL VOC2012 SEGMENTATION DATASET

Methods	Bcg.	Aero.	Bicy.	Bird	Boat	Bottle	Bus	Car	Cat	Chair	Cow	D.T	Dog	Horse	M.B	Per.	P.P	Sheep	Sofa	Train	TV	mIoU
MCT+ [19]	87.39	**64.85**	33.52	71.79	58.95	61.05	76.71	68.67	79.93	31.62	72.18	47.99	76.88	72.16	68.67	67.51	**50.85**	67.63	50.34	70.31	**59.21**	63.72
Baseline	85.37	59.33	**37.02**	66.39	46.76	55.46	64.93	65.54	**81.36**	**32.83**	72.76	42.31	76.76	71.95	**71.01**	61.05	50.34	71.11	50.58	57.99	53.34	60.68
Base+CON	87.23	66.28	32.95	70.41	60.64	55.50	73.87	70.39	77.03	27.94	72.58	47.43	76.17	73.53	66.21	68.75	48.73	67.70	50.65	69.75	56.86	62.89
Base+SAL	86.98	64.46	31.83	71.44	56.36	58.21	75.51	**72.01**	80.74	30.10	69.77	**51.03**	**77.49**	73.64	68.30	66.30	49.72	68.89	48.66	**72.04**	52.91	63.21
STAC	**87.70**	68.18	33.54	**71.68**	**62.86**	**61.58**	**77.77**	71.32	78.11	29.82	**75.11**	47.38	76.70	**73.93**	67.12	**67.65**	50.56	**71.31**	**53.40**	70.59	57.89	**64.48**

N.B.: All results are reported based on CAM seeds without applying any post-processing methods, including those for MCT+ [19].
The bold values indicate the best values.

TABLE III
PERFORMANCE COMPARISON OF THE PROPOSED APPROACH VARIANTS ON THE MVTEC ANOMALY DETECTION DATASET

Category	Base-CNN			Base-T			STAC-CNN			STAC-T		
	mIoU	FP	FN	mIoU	FP	FN	mIoU	FP	FN	mIoU	FP	FN
Bottle	62.02	27.74	10.25	65.62	11.98	22.40	69.33	16.56	14.11	**88.04**	5.48	6.48
Carpet	56.67	38.99	4.34	71.08	6.21	22.7	69.75	19.88	10.38	**79.15**	6.69	14.16
Grid	44.24	47.29	8.47	60.50	22.97	16.52	61.14	25.74	13.12	**63.27**	21.69	15.04
Hazelnut	56.57	36.15	7.28	69.65	23.13	7.22	65.10	21.57	13.32	**86.49**	7.64	5.89
Leather	50.34	45.25	4.41	68.05	17.52	14.42	68.01	24.78	7.21	**74.09**	13.74	12.17

TABLE IV
SEGMENTATION RESULTS ON THE NEU-SEG DATASET

Models	Backbone	Bg.	Inc.	Pat.	Scr.	mean
Full	–	97.59	76.05	88.37	81.61	85.90
WDLS [8]*	ResNet-FCN8s	–	–	–	–	69.77
PLDL [10]*	ResNet-FCN8s	–	–	–	–	74.33
Using the refined CAM from **STAC**						
Wide-ResNet	Res38	94.86	53.07	77.71	63.73	72.30
DLV3+	ResNet18	95.53	52.29	77.78	69.47	73.80
DLV3+	EffNet-B0	96.17	61.58	82.25	72.07	78.00
DLV3+	**EffNet-B4**	**96.25**	**62.63**	**82.84**	**72.03**	**78.40**
SegFormer	MiT-B0	96.11	61.42	82.39	70.69	77.60
SegFormer	MiT-B4	96.18	61.84	82.39	71.96	78.10

* indicates the results are taken as reported in this article.

TABLE V
SEGMENTATION RESULTS (MIOU) ON THE DAGM AND MT DATASETS

Methods	Backbone	DAGM		MTD	
		Train	Test	Train	Test
Wide-ResNet	Res38	56.60	56.00	48.20	49.10
DLV3+	ResNet18	51.20	52.4	64.00	64.9
DLV3+	EffNet-B0	65.40	**66.40**	64.20	64.40
DLV3+	EffNet-B4	**65.80**	65.90	68.20	68.6
SegFormer	MiT-B0	62.40	63.10	69.10	**68.90**
SegFormer	MiT-B4	64.50	65.30	**69.20**	68.50

The bold values indicate the best values.

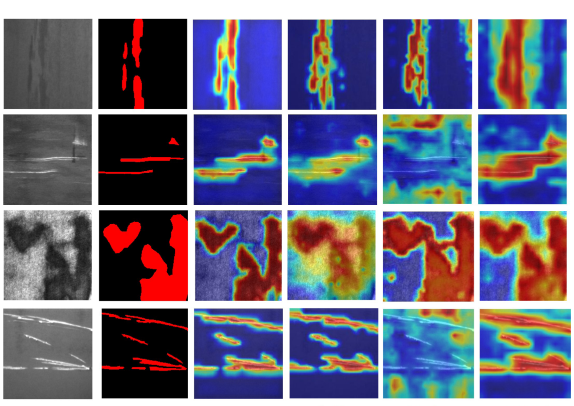

Fig. 4. Illustration of localization results of CAM on samples from the NEU-Seg.

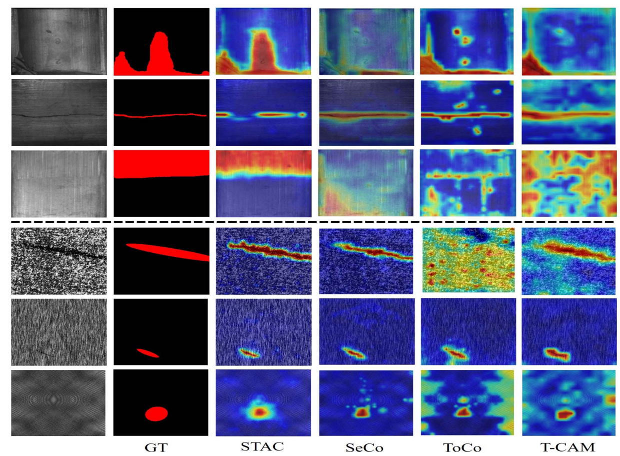

Fig. 5. Comparison of the localization map generated by STAC and other methods on MTD dataset (top) and DAGM dataset (bottom).

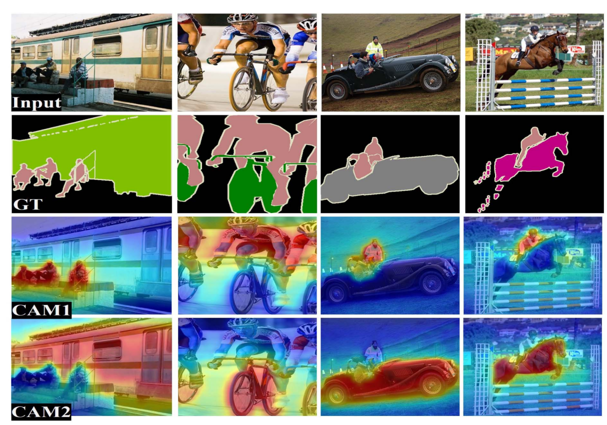

Fig. 6. Qualitative results on the PASCAL VOC2012 dataset. The selected visualizations demonstrate the generalization capability of the proposed approach across different domains.

the CNN-based approach yields less precise results with larger FP regions. This indicates that the CNN-based model tends to falsely activate nondefective areas, further validating the motivation and effectiveness of the proposed approach .

4) Visualization of Failure Cases: We provide detailed visualizations of failure cases to show and analyze the limitation of the proposed method. As depicted in Fig. 8, our approach exhibits limitations in precisely localizing defect regions, particularly in scenarios characterized by low contrast, significant intraclass variability, and indistinct boundaries. These instances exemplify practical challenges frequently encountered in industrial defect detection datasets. Nevertheless, despite these

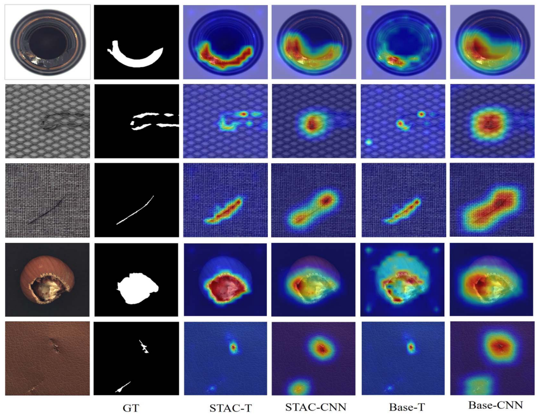

Fig. 7. Visualization of results on the MVTec dataset.

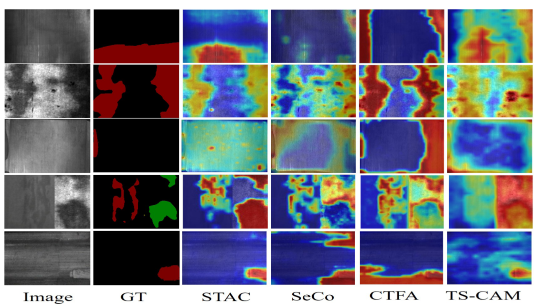

Fig. 8. Comparison of failure cases with different approaches.

TABLE VI
EFFECTS OF DIFFERENT MODEL COMPONENTS IMPLEMENTED ON BOTH CNN-BASED AND TRANSFORMER-BASED ATTENTION

	Class-IoU				Overall		
Modules	Bg.	Inc.	Pat.	Scr.	mIoU	FP	FN
CNN-base	61.52	16.30	35.89	17.85	32.89	55.64	11.47
+CONT	77.55	17.40	46.03	18.06	39.76	41.66	18.58
+SAL	91.33	29.69	68.82	32.68	55.63	20.48	23.88
Ours-CNN	93.99	42.84	78.90	55.78	67.88	17.82	14.29
Tran-base	81.90	19.42	9.34	39.96	37.66	32.71	29.64
+PT	80.03	21.19	30.26	39.21	42.67	42.48	14.84
+CONT	85.68	30.91	13.45	44.79	43.71	29.61	26.69
+SAL	95.28	54.10	80.00	61.75	72.78	11.66	15.56
STAC	**95.39**	**58.87**	**81.69**	**62.90**	**74.71**	**14.46**	**10.83**

The bold values indicate the best values.

complexities, our method consistently demonstrates marginally superior localization performance relative to other approaches. In addition, it is worth noting that although the proposed method achieves superior localization performance compared to the other approaches, certain challenging classes, such as Inclusion in the NEU-Seg dataset and Blowhole, Crack, and Uneven in the MTD dataset, remain difficult, leading to higher FP and FN rates, as given in Tables III, VI, and VII, and Fig. 8 .

E. Ablation Study

1) Effects of Model Components and Architectures: Three main components of the model are presented in the proposed approach (see Fig. 3). The baseline method is implemented only with classification task, then the *PT* indicates the inclusion

TABLE VII
EFFECTS OF CAM THRESHOLD (T)

	Class IoU				Overall		
T	Bg.	Inc.	Pat.	Scr.	mean	FP	FN
0.4	92.05	47.71	79.04	49.61	67.10	28.63	4.26
0.5	94.35	55.99	**82.09**	57.54	72.49	21.32	6.19
0.6	**95.39**	**58.87**	81.69	**62.90**	**74.71**	14.46	10.83
0.7	95.24	53.97	76.73	60.61	71.64	8.62	19.74
0.8	93.91	38.37	66.28	46.98	61.39	4.80	33.83

The bold values indicate the best values.

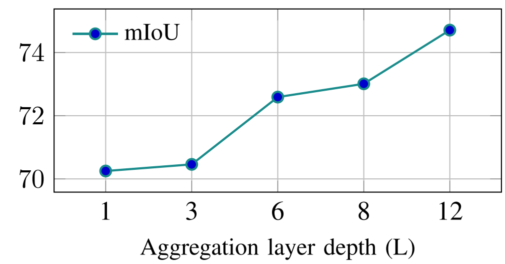

Fig. 9. Effects aggregating attention map from different Transformer layers (L) on mIoU. Increasing depth improves mIoU up to $L = 12$.

of auxiliary attention refinement module (PEM), *CONT* is the contrastive learning module, and *SAL* is the saliency-guided refinement module. To further compare with the performance of CAMs generated by CNN-based methods, we additionally implemented our approach using the widely adopted ResNet38 [44] backbone. As presented in Table VI, each component of the proposed method contributes to a notable improvement in performance, with the combination of all modules yielding the best overall results. Moreover, the Transformer-based implementation significantly outperforms its CNN-based counterpart, supporting our motivation that Transformer attention maps enable the model to more effectively capture global contextual relationships.

2) Effects of Attention Weight Depth and Layer Index: The DEiT Transformer is implemented with a depth of 12 MHA layers, designed to capture complex hierarchical representations. However, during model evaluation, aggregating attention maps from all these layers may not significantly contribute to performance, as redundant or overlapping information might be encoded across the layers. To address this, we systematically investigated the impact of aggregating attention maps from specific layer depths, aiming to identify the optimal combination for enhancing model performance. The results of this analysis, highlighting the contribution of different depths, are presented in Fig. 9. As shown in the figure, the aggregation from all layers improves the localization capability of the CAM improving the mIoU results. This indicate that aggregating attention maps from deep to shallow layers, high-level semantics and low-level spatial detail, improves robustness across diverse defect types.

3) Effects of CAM Threshold: We applied threshold (T) to the computed CAM to distinguish between background and foreground predictions during inference. As given in Table VII, the choice of threshold values significantly impacts performance. It was observed that lower thresholds (e.g., 0.4) tend to introduce FP, particularly in background regions, due to the inclusion of low-confidence activations. Higher thresholds (e.g.,

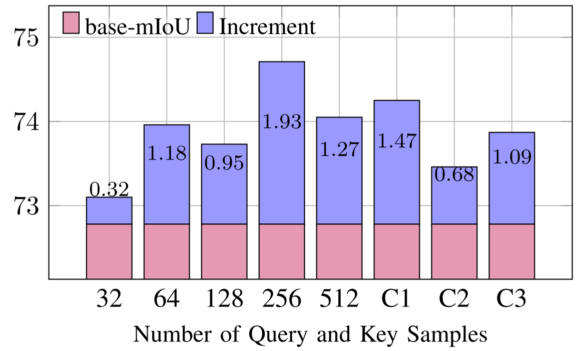

Fig. 10. Effect of Query and Key sampling strategy. Equal sampling is denoted by single values (e.g., 32), while unequal sampling configurations are labeled as C1 (256–128), C2 (128–256), and C3 (512–256), where the first value indicates the number of query samples and the second the number of key samples.

0.8) were found to result in FN, omitting valid but low-activation defect regions, especially for low-contrast defects, such as Inclusion. No consistent class-specific advantages were noted with varying thresholds, and a threshold of $T = 0.6$ was determined to offer the best tradeoff across all classes.

4) Effects of Query and Key Sampling Strategy: We evaluated two main strategies to select representative positive and negative pixel pairs during training: Equal sampling with a fixed number of positive and negative samples (e.g., 128 each), and unequal sampling, where positive and negative samples are drawn differently (denoted as strategy C in Fig. 10). We conducted an ablation study to compare these strategies, and the results show that equal sampling with 256 positive and negative pairs consistently outperforms other configurations. Based on this finding, all reported experiments in this article use this sampling strategy.

5) Effect of Loss Balancing Coefficients (λ and γ): This section analyzes the impact of individual loss components and their corresponding weighting coefficients on model performance. As illustrated in Fig. 11, training stability and effectiveness are highly sensitive to the choice of these balancing coefficients. Specifically, lower values of the contrastive loss coefficient (γ) combined with higher values of the saliency consistency loss coefficient (λ) lead to improved performance. Based on these observations, we set $\lambda = 2.0$ and $\gamma = 0.2$ for all reported experimental results.

6) Backbone Networks and Complexity Analysis: We perform a complexity analysis on both the CAM generation networks and the segmentation networks using different backbone networks. The analysis is done using NVIDIA RTX 4090, batch size = 1, and input resolution of 224×224. We apply "torch.cuda.max_memory_allocated" for all models to determine peak memory allocation during inference. As given in Table VIII, our method has significantly lower parameters and computational costs. STAC employs a lightweight transformer backbone (DeiT-S), achieving high inference speed, well above the commonly required 30 FPS threshold for real-time industrial applications. However, these results are based on high-end hardware, and performance may vary depending on the deployment environment. Moreover, generated CAMs are used to supervise segmentation models as a pseudopixel-level label,

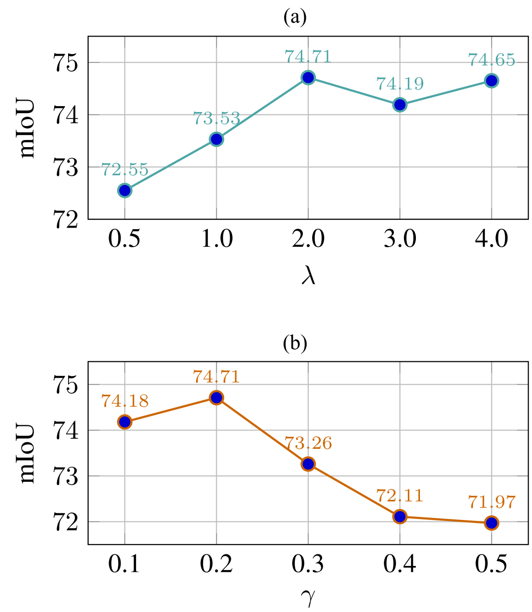

Fig. 11. Ablation study on coefficients λ and γ, showing their influence on model CAM performance (mIoU). (a) Effect of λ. (b) Effect of γ.

TABLE VIII
COMPLEXITY ANALYSIS ON INPUT SIZE OF 224×224

Model	Backbone	#Par.(M)	MACs(G)	FPS	Memory (MB)
STAC	DeiT-S	**21.75**	**4.29**	**283**	**156.03**
T-CAM	Conformer-S	36.33	10.62	121	303.71
EPS	Res38	105.09	99.90	102	554.04
		Segmentation models			
Res38	Res38	123.96	114.71	153	1029.60
DLV3+	ResNet18	12.33	3.52	**516**	195.88
DLV3+	EffNet-B0	4.91	**0.45**	246	164.84
DLV3+	EffNet-B4	18.62	0.54	155	218.24
SegFormer	MiT-B0	**3.45**	0.65	372	**164.59**
SegFormer	MiT-B4	61.04	8.90	95	386.48

The bold values indicate the best values.

including lightweight CNNs and Transformer-based models, as given in Tables IV and V. The analysis reveals well-balanced performance-speed tradeoffs. For example, DeepLabV3+ with a ResNet18 backbone achieves 73.8 mIoU at ∼516 FPS.

V. CONCLUSION

In this work, we propose a novel weakly supervised approach for defect localization and segmentation, addressing key challenges in intelligent industrial inspection. Our method leverages saliency-guided, class-specific CAMs derived from Transformer attention, combined with contrastive learning, to generate precise localization maps. A major challenge in this task is the presence of inconsistent defect boundaries, interclass similarity, and intraclass variation, which make it difficult to accurately localize defects using CAMs generated from a classification model. To address this, we refine the CAMs using saliency maps to enhance foreground and background learning. The refined CAMs further guide pixel-level contrastive learning by generating positive and negative pairs in the feature space, improving the model's ability to distinguish between different defect classes in complex industrial images. Extensive experiments and ablation studies demonstrate that our approach significantly outperforms state-of-the-art methods. This research contributes to advancing weakly supervised learning techniques and enhancing deep learning-based solutions for industrial inspection, particularly in scenarios where annotated data is scarce and expensive to obtain. Future work could focus on integrating the proposed method with complementary techniques, such as deep level-set frameworks or multimodal learning, to further enhance its performance and representation capability. Moreover, extending the approach toward domain generalization and incremental learning would enable it to adapt to newly emerging defect types and maintain robustness against label noise, thereby improving its scalability and practicality for real-world industrial applications.

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